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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: EET 305

Course Name: SIGNALS AND SYSTEMS

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|---|---|
| 1 | Plot $x(2t)$, if $x(t) = u(-t + 2) + u(t - 3)$. | 3 |
| 2 | Find whether following signal is odd or even, $x(t) = t^4 + 4t^2 + 6$. | 3 |
| 3 | State and prove the time shifting property of Fourier Transform. | 3 |
| 4 | Derive the electrical analogous elements corresponding to elements of mechanical translational system with force voltage analogy. | 3 |
| 5 | Explain the concept of positive real functions. | 3 |
| 6 | Determine the impulse response of the system having transfer function,
$H(s) = \frac{s}{(s-2)(s+3)}$ | 3 |
| 7 | Define the process of sampling and state sampling theorem. | 3 |
| 8 | Write any three properties of z-transform. | 3 |
| 9 | Check the stability of the system represented by the following characteristic equation using Jury's test: $2z^4 + 7z^3 + 10z^2 + 4z + 1 = 0$ | 3 |
| 10 | Obtain the transfer function of given discrete time system described by the difference equation: $y(n) - \frac{5}{4}y(n-1) + \frac{1}{6}y(n-2) = 2x(n)$ | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module-1**

- | | | |
|-------|--|---|
| 11 a) | Find the response of a system with an impulse response of $h(t) = u(t)$ for an input signal of $x(t) = e^{-\alpha t} u(t)$ | 8 |
| b) | Determine whether the following systems are time-invariant and linear. | 6 |
| (i) | $3 \frac{dy(t)}{dt} + 5ty(t) = x(t)$ | |
| (ii) | $\frac{dy(t)}{dt} + 10 \sin y(t) = 2x(t)$ | |
| (iii) | $y(t) \frac{dy(t)}{dt} + 10 y(t) = 2x(t)$ | |

- 12 a) For the signal, $x(t)$ shown in Fig.1, plot: (i) $x(t - 1)$ (ii) $x(-3t + 2)$

8

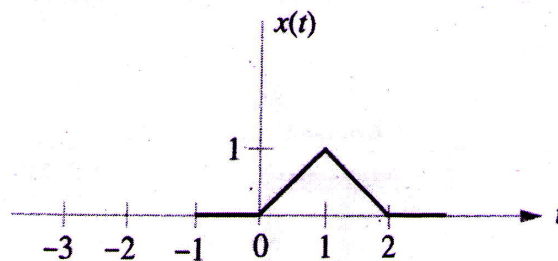


Fig.1

- b) Test whether the following signals are periodic or not. If periodic, find the fundamental period.

6

(i) $x(t) = 5 \sin 24\pi t + 7 \sin 36\pi t$

(ii) $x(t) = e^{j(\pi t - 2)}$

(iii) $x(t) = 5 \sin 24\pi t \cos 36\pi t$

Module -2

- 13 a) Determine the transfer function, $\frac{V_C(s)}{V_S(s)}$ of the given circuit. Hence, find the impulse response of the circuit.

4

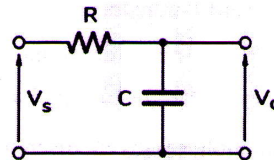


Fig.2

- b) Obtain the trigonometric Fourier Series of the given signal

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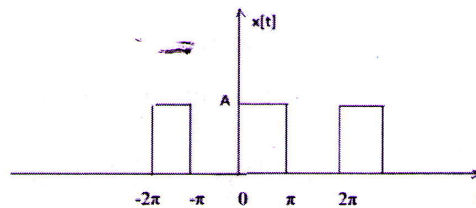


Fig.3

- 14 a) Derive the transfer function $\frac{X_2(s)}{F(s)}$ of the given mechanical system in Fig.4

10

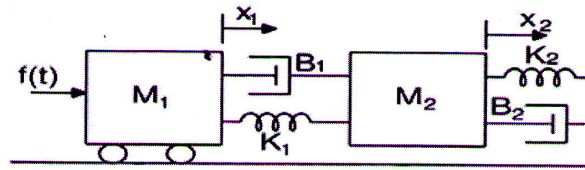


Fig.4

- b) Find the transfer function of given LTI system.

4

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 2\frac{dx(t)}{dt} - 3x(t)$$

Module -3

- 15 a) How time response can be obtained from pole zero plot. 4
 b) Describe Mason's gain formula for signal flow graph. Obtain the transfer function of the following system using Mason's gain formula. 10

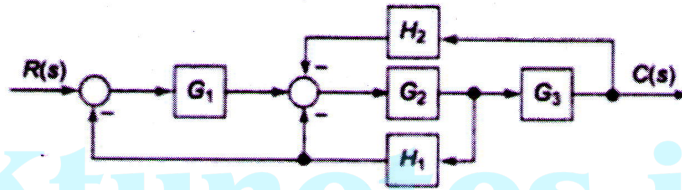


Fig.5

- 16 a) Check if the system with the given characteristic equation is stable using Routh Hurwitz stability criterion, $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$ 7
 b) Obtain the transfer function for the following system using block diagram reduction technique. 7

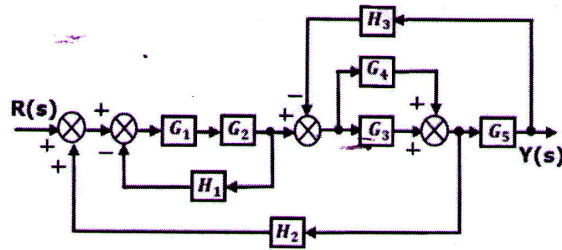


Fig.6

Module -4

- 17 a) Explain the significance of Zero order hold circuit in signal reconstruction. 4
 b) Find the inverse Z-transform of: 10

$$X(Z) = \frac{z^{-1}}{3 - 4z^{-1} + z^{-2}}; \text{ROC} |Z| > 1$$

- 18 a) List any four properties of Discrete convolution. 4
 b) Find the convolution sum of the following sequences 10
- $$x(n) = 3 \delta(n+1) - \delta(n) + 2 \delta(n-1) + \delta(n-2)$$
- $$h(n) = 2 \delta(n) - 2 \delta(n-1) + 2 \delta(n-2) + \delta(n-3)$$

Module -5

- 19 a) State and prove the time reversal property of Discrete time Fourier transform. 4
 b) A discrete system is given by the following difference equation: 10
- $$y[n] - 5 y[n-1] = x[n] + 4 x[n-1]$$
- where, $x[n]$ is the input and $y[n]$ is the output. Determine the expression for its magnitude response and phase response.

- 20 a) Check the stability of the given system using bilinear transformation: 6
- $$F(z) = z^3 - 0.2z^2 - 0.25z + 0.05 = 0$$
- b) Obtain the direct form-I and direct form-II realizations of the discrete time system with system function as: 8

$$H(z) = \frac{8 - 4Z^{-1} + 11Z^{-2} - 2Z^{-3}}{1 - \frac{5}{4}Z^{-1} + \frac{3}{4}Z^{-2} - \frac{1}{8}Z^{-3}}$$

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